Aerospace Manufacturing Core
Study Guide

"AMCSG-205"

Tools & Processes

Composites

Composites

Text from ChatGPT

Composite Types

- **Composites** beat metal because of
  - Higher **Strength-to-Weight** ratio
  - Better **Corrosion** and **Fatigue Resistance**
  - More Complex, **Aerodynamic** Shapes
  - **Fewer Parts**
- Three Types:
  - **Fiberglass**
  - **Aramid** (Kevlar)
  - **Carbon Fiber**
- **Made of Fiber + Resin + Core**

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Carbon Fiber

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Aramid

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Fiberglass

- ChatGPT: “make one picture showing three types of aircraft manufacturing fiber side by side: fiberglass, aramid, and carbon fiber.”

Text from ChatGPT

Composite Type Strengths

Fiberglass

- Resistant to Harshest Chemicals
- Least Expensive

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Carbon Fiber (Graphite)

- Lowest Density / Highest Strength
- Low Thermal Expansion

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Aramid

- Good Impact & Abrasion Resistance
- Flame / Thermal Degradation Resistant
- Chemical Resistant

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- Fiberglass – ChatGPT: make an image indicating fiberglass is the least expensive, and is resistant to chemicals + Edit output manually
- Aramid – ChatGPT: make an image indicating Aramid is flame/thermal resistant, chemical resistant, and as kevlar is impact and abrasion resistant
- Carbon Fiber – ChatGPT: make an image indicating carbon fiber has low thermal expansion, and low density/high strength

Text from Google AI

Carbon Fiber Weave Patterns

- **Plain Weave**
  - Simple over/under checkerboard
  - Flat, Structural, or Interior Cabin
- **Twill Weave**
  - Highly Versatile, Complex Contours
- **Satin Weave**
  - Highly Pliable, Complex Contours
- **Unidirectional Weave**
  - Tape, not a weave / one direction
  - Wings, spar caps

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- ChatGPT: “make a picture showing the various Carbon fiber weaves in aircraft manufacturing. Show plain, twill, satin, and unidirectional weaves.”

- Google AI: “what are the different weave patterns for carbon fiber for aircraft manufacturing”

Text from Google AI

Resins

- Final Composite Material = Fiber + Resin
  - Resin hardens the fiber into final shape
    – "without Resin…it's just a pile of floppy strings ☺"
- Four Types of Resin:
  - **Thermoset** resin systems – plastic, needs to be heated to cure, good for aircraft
  - **Phenolic** resin systems – fire retardant, heat resistant, good for aircraft
  - **Pre-Impregnated** materials – like a roll of tape with resin already embedded
  - **Epoxy** resin systems – widely used for adhesion, strength, resist chemicals

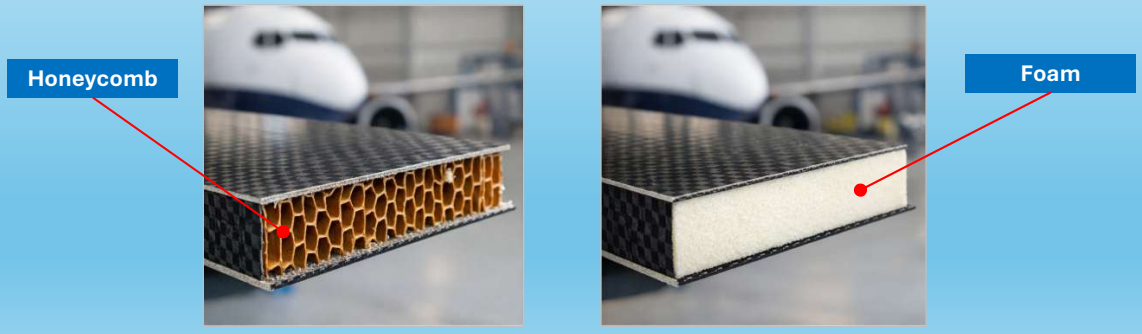
- Google AI: "what are the different weave patterns for carbon fiber for aircraft manufacturing" … Prior slide research…talked about resin, so I added a slide here too

Text from ChatGPT

Core Types

A “Core” is Material Sandwiched Between 2 Composite Sheets

- **Honeycomb** Core – Aluminum, Nomex, Carbon Fiber, Phenolic
- **Foam** Core – Styrofoam, Urethane, PVC



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ChatGPT / Matt Pierce, 2026

- ChatGPT: “make a side-by-side picture showing two different core materials (honeycomb vs. foam) for aircraft composite materials”

Text from GoogleAI

Layup Process

The "**Layup Process**" is simply stacking layers of carbon fiber onto a mold to build a part's shape.

1. **Preparation**: Mold cleaned then coated with release agent
2. **Material Application**: Layers of composite material added to mold
3. **Orientation/Debulking**: Layer orientation varies for strength. Debulking is compacting to remove air.
4. **Core Materials**: Add honeycomb or foam core between layers if req'd

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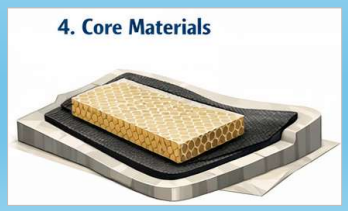

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- Google AI: "simple definitions for "layup process" and "bagging process" in aircraft manufacturing of composite parts"
- Google AI: "what is the layup and bagging process in aircraft manufacturing of composite parts"
- ChatGPT: "Make a step by step process diagram showing how the layup and bagging process is done in aircraft manufacturing of composite parts. Show "Layup Process" as four steps: Preparation, Material Application, Orientation/Debulking, and core materials. Show "Bagging Process" as four steps: Peel Ply/Release Film, Breather Material, Vacuum Bag / Seal, and Vacuum Application/Cure."

Text from GoogleAI

Bagging Process

The "**Bagging Process"** is a technique to seal the layered, uncured composite part inside a plastic airtight bag, then removes air for uniform pressure during curing.

1. **Peel Ply & Release Film**: Placed over uncured part to prevent bag from sticking
2. **Breather Material**: Porous breather cloth added facilitate uniform vacuum
3. **Vacuum Bag & Seal**: Flexible, airtight, nylon bag cover entire assembly; sealed to mold with sealant tape
4. **Vacuum Application & Cure**: Pump out air, atmospheric pressure consolidates layers + resin. Cured in autoclave or oven under controlled heat.

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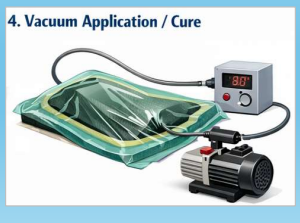

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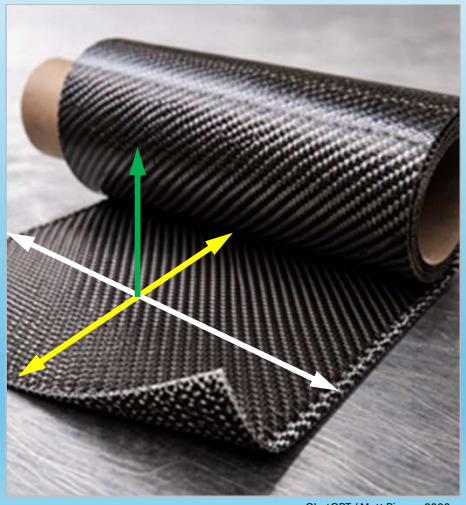
Text from GoogleAI

Fiber “Warp Threads” and “Fill Threads”

- **Warp Fibers** – “Yellow Arrow”
  - Runs Length of Roll
  - = “Warp Direction”
  - Strongest in this direction
- **Warp Face** – “Green Arrow”
  - “Up” = Top Face as is Unrolled
- **Fill Threads** – “White Arrow”
  - Perpendicular to Warp fibers
  - Parallel/Lateral to Roll



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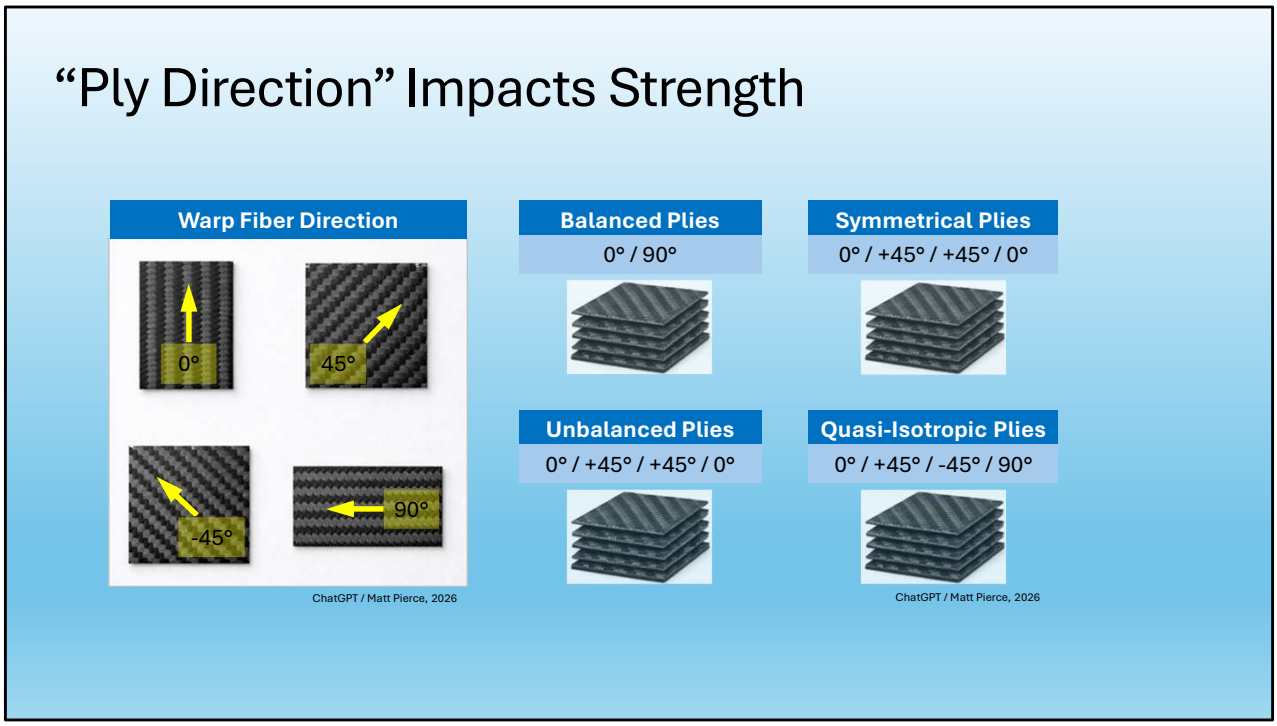
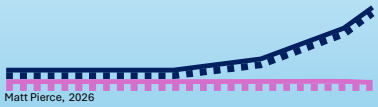
- Google AI: “composite material fiber, ply orientation, warp direction, warp face, warp threads, fill threads, warp clock”
- Chat GPT: “make a single diagram showing a roll of composite fiber material coming off the roll; be zoomed in on the material as it comes off the roll so can see the warp face and warp direction”
  + Remove the labels and arrows for "warp face" and "warp direction“
  + make the unrolled surface curled up a bit (not flat), and cut it about 12" from the end



- Chat GPT: “make a photo showing 4 small 2" x 2" squares of carbon fiber. show one square at warp ply direction 0°, one at -45°, one at 45°, and one at 0°.”
  + “make the upper left square go up-and-down vertically. make the lower right square label = 90° and direction be left-to-right horizontally.”
  + “Remove text labels. Leave pictures as they are.”
- Chat GPT: “make a diagram showing the following: 1. a small square of composite fabric with its ply orientation = warp fiber direction. 2. several small squares of composite fabric with a "Balanced" ply orientation 3. several small squares of composite fabric with an "Unbalanced" ply orientation 4. several small squares of composite fabric with a "Symmetrical" ply orientation 5. several small squares of composite fabric with a "Quasi-Isotropic" ply orientation”

“Ply Orientation” Impacts Strength

Stacking



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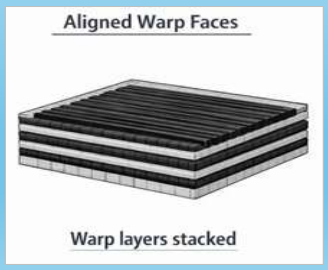

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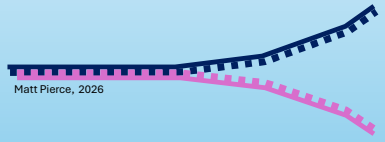
Nesting



Matt Pierce, 2026

ChatGPT / Matt Pierce, 2026

- Chat GPT: make a side-by-side diagram showing the two types of orienting warp faces together of composite fabric sheets: "nesting" vs "stacking“

Q&A

Practice Q&A's (Q1-Q8)

Q1: Composites beat metal because they have higher strength to weight?

A: True

Q2: Composites beat metal because they do not corrode?

A: True

Q3: Composites beat metal because they are more fatigue resistant (bend until failure)?

A: True

Q4: Composites beat metal because they can be molded int more complex aerodynamic shapes?

A: True

Q5: Composites beat metal because there are fewer parts to fasten together?

A: True

Q6: Composites beat metal because they are cheaper to make?

A: False

Q7: What is the oldest composite that has been in wide use?

A: Fiberglass

Q8: What is the strongest composite for impact (like stopping a bullet)?

A: Aramid fiber (Kevlar)

Practice Q&A's (Q9-Q16)

Q9: What is the strongest composite for aerospace manufacturing?

A: Carbon fiber

Q10: What composite is the most commonly used today?

A: Fiberglass (like your bathtub wall)

Q11: What composite is best for flame and heat resistance?

A: Aramid fiber

Q12: What fiber has low thermal expansion?

A: Carbon fiber (not stretch like Aluminum on hot days)

Q13: Name of string patterns in the fiber plies (e.g.: plain, twill, satin, or unidirectional)?

A: Weave

Q14: Component of a composite that hardens the soft fiber plies into final shapes?

A: Resin

Q15: Name four types of resin?

A: 1. Thermoset, 2. Phenolic, 3. Pre-Impregnated, and 4. Epoxy

Q16: Name two common core types sandwiched between two outer layers of composite faces

A: 1. Honeycomb, or 2. Foam

Practice Q&A's (Q17-Q23)

Q17: Part of the composite process where stacking layers of carbon fiber over a molded?

A: Layup Process

Q18: Part of the composite process to seal mold and material placed during Layup Process?

A: Bagging Process

Q19: Is there a bag used in the Bagging Process?

A: No! It is a clear plastic sheet taped down to make an airtight seal over the mold and material that forms a "bag"

Q20: Name of threads in a ply that run longitudinally (runs length of the roll) and are strongest?

A: Warp threads

Q21: Name of threads in a ply that run perpendicular to warp fibers, laterally across the roll?

A: Fill threads

Q22: Name of the face that curls up when you unroll composite sheet?

A: Warp face

Q23:Name of "schedule" indicating how to layer the warp directions of ply after ply when building a parts?

A: Warp clock

Practice Q&A's (Q24-Q30)

Q24: Name of ply orientations where warp faces are aligned?

A: Stacking

Q25: Name of ply orietnations where warp faces interlock together?

A: Nesting

Q26: Using honeycomb or foam cores serves what purpose?

A: Increase strength

Q27: What is the most widely used resin?

A: Epoxy

Q28: Name of material used to allow air and fuems to outflow during vacuum and curing processes?

A: Breather material

Q29: Name of cloth fabric used to absorbe excess resin during vacuum and curing processes?

A: Bleeder blanket

Q30: What is the purpose of quasi-isotropic ply orientations?

A: Strength in multiple directions (flexing, twisting)

Thank You

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https://github.com/DataResearchLabs/study_guide_intro_aerospace_mfg